Biased-noise thresholds of cubic and rhombic codes: a whole-dissertation replication audit of Pesah (2021)

30 October 2026

Abstract

We did not replicate the reported cubic pure-Z threshold improvement under Hadamard deformation: the published BP–OSD thresholds are 21.7% undeformed and greater than 36% deformed, whereas we estimate 21.908% undeformed and cannot identify a reliable deformed threshold. Unknown original Monte Carlo error, missing production settings and inadequate finite-size scaling fits prevent an exact comparison; this failure to replicate does not refute the reported improvement. For cubic Sweep–Match under depolarizing noise, the deformation contrast reverses from +0.300 to −0.134 percentage points, with a different matching engine; an unstable fit also reverses a cross-code point ranking at Z bias 100. Supporting results recover the direction of cubic pure-X improvement (2.9%/7.4% published versus 2.388%/7.570%) and rhombic pure-Z deterioration (28.4%/6.5% versus 27.407%/6.490%). Independent numerical checks and analytical arguments verify code parameters, ideal-majority limits and conditional decoder guarantees. These findings support channel-dependent benefits of deformation, but do not establish the principal cubic pure-Z improvement or the actual BP–OSD asymptotic endpoint.

Claim	Published	Ours	Same direction?	Verdict
Cubic Z, BP–OSD	21.7 / > 36	21.908 / unresolved	Unknown	No match
Cubic X, BP–OSD	2.9 / 7.4	2.388 / 7.570	Yes	Partial
Rhombic Z, BP–OSD	28.4 / 6.5	27.407 / 6.490	Yes	Partial
Cubic dep., Sweep–Match	4.4 / 4.7	4.894 / 4.759	No	No match

Threshold pairs are undeformed/deformed percentages from Tables 3.1–3.2 of Pesah (2021) and our fits. Positive deformed-minus-undeformed favors deformation. Headline verdicts concern the complete comparison, not statistical equivalence. Numerical matching requires rounding half up to the published 0.1 percentage point; a gap of at most one last-place unit is *close*. Unknown source uncertainty and unreliable fits limit inference; the unresolved deformed threshold has no numerical agreement verdict.

1 Introduction

We did not replicate the reported cubic pure-Z improvement from 21.7% to greater than 36%: our undeformed estimate is 21.908%, but the deformed threshold is not reliably identified. Missing production information and failed scaling diagnostics limit this comparison. The Sweep–Match depolarizing contrast changes from +0.300 to −0.134 percentage points, and an unstable cubic fit reverses a cross-code point ordering at Z bias 100. These are point-estimate discrepancies, not evidence of a statistically established

opposite ordering. Numerical agreement uses the 0.1-percentage-point rule stated above, separately from uncertainty and reconstruction scope.

The scientific question is whether local Hadamard deformation improves a code's tolerance of biased Pauli noise without changing its physical-qubit count. We investigate the numerical, analytical and software results in Arthur Pesah's *Tailoring 3D topological codes and decoders to biased noise*, a University College London Master of Research dissertation submitted on 1 September 2021 under Dan Browne (Pesah, 2021). Its contributions are the code deformations, decoder implementation and published comparisons. Our contributions are reconstructed experiments, independent implementation checks and analytical verification of the stated reductions and code parameters.

The evidence supports different effects in different channels. Cubic pure-X improves in the reconstruction, whereas rhombic pure-Z worsens strongly. Agreement in these cases, or equality of code resources, does not establish the unverified cubic pure-Z improvement. The study covers endpoint thresholds, finite-bias comparisons, total/point/loop curves, code constructions, the OSD optimization and the graphical interface. The 87-item `coverage.json` distinguishes assessed findings and access limits from inherited background and proposed future experiments.

2 Experimental reconstruction

We run periodic sizes $L = 4, 6, 8, 10$, both deformation arms, perfect syndrome measurements and independent physical-qubit errors. Categorical I/X/Y/Z draws preserve the simultaneous X and Z components of Y. For bias η along one Pauli axis, its probability is $p\eta/(\eta + 1)$ and each other axis receives $p/[2(\eta + 1)]$. Depolarizing noise is $\eta = 1/2$; the pure channels are the limiting cases. Hadamard deformation swaps X and Z on the selected axis.

The primary decoder combines ten iterations of belief propagation (BP) with order-zero ordered-statistics decoding (OSD), reconstructing the implementation in Huang et al. (2021); its algorithmic context is Roffe et al. (2020). The source does not identify production seeds, counts, iteration settings, package versions, tie handling or fitting windows. Current NumPy quicksort, the historical prior regularizer and independent PCG64/MT19937 trials are explicit supplied assumptions. Full decoder equivalence to the undisclosed source production run is not established.

An original C++ GF(2) elimination routine accelerates OSD while preserving reliability ordering, pivot selection and zero free variables. It matched the Python reference routine in 196 correction cases through $L = 10$. The generalized cubic decoder matched 480 historical dense-reference cases; the rhombic implementation matched 320, after an explicit global Pauli conversion. These checks share graphs, priors and the stated algorithm. They independently check arithmetic/correction implementations, not an entire independent scientific replication.

The rhombic convention is consequential: the thesis prose uses cube Z checks and triangle X checks; the available historical code globally interchanges them. We use the prose convention and verify the exact global-Hadamard mapping. A nearby figure caption also conflicts with the prose. Table 3.2 calls the lattice cubic although its surrounding section and values describe rhombic. Figure 3.6 labels the undeformed pure-Z endpoint 21.0 percent versus Table 3.1's 21.7 percent; Figure 3.9 labels deformed depolarizing 1.6 percent versus Table 3.2's 1.7 percent. We retain table values for endpoint targets and record the figure discrepancies, rather than silently selecting whichever agrees best. This choice explains our channel labeling but does not prove which code revision generated the published plots.

Historical Sweep code is executed externally with PyMatching 2.3.1. The original 0.2.4 build required a LEMON archive whose host denied access. The modern sparse-blossom implementation (Higgott and Gidney, 2025) is a disclosed engine departure. Eighty finite matching checks agree with independent

exhaustive pairing of up to six defects, using shared shortest-path distances and graphs; they do not prove historical tie equivalence. All nonzero residual syndromes count as failures: 2,696 of 19,200 Sweep trials retain a residual syndrome. These are retained decoder failures, not discarded trials; all BP production corrections satisfy their syndromes.

The pure-Z grid contains 8,800 fresh trials and took approximately 2,478 seconds. The 28 BP families contain 179,200 trials; three Sweep families contain 19,200. Three wider diagnostic grids add 38,400. A further 5,600 shared-error diagnostic trials compare ten versus 100 BP iterations and stable versus quicksort ties (16,800 decoder executions). The full source-vertex curve run contains 226,800 trials; these production and wider-grid runs total 472,400. Individual production processes are capped at 2 GiB address space and 2,400 CPU seconds, with bounded worker concurrency. The cubic pure-X threshold experiment uses a separate 16,000-trial sample; validation and pilot runs are not included in these totals. Output files record measured times, seeds and completion flags. All production files passed completion, row-cardinality and count-integrity checks. Across separately reported families, 406 generator/seed keys are reused; those families are not pooled as independent evidence. The six cross-code contrasts use disjoint simulation seeds. Bonferroni requires no cross-test independence.

3 Numerical agreement and uncertainty

The endpoint comparisons recover some effect directions but do not establish numerical agreement across the reported thresholds. The following table covers every endpoint cell of source Tables 3.1–3.2. U and D denote undeformed and deformed codes; positive $D - U$ favors deformation. The deformed cubic Z source is a lower-bound statement, not an exact 36% measurement.

Code/decoder	Noise	Source (%)		Ours (%)		$D-U$ (pp)	
		U	D	U	D	Source	Ours
Cubic BP	X	2.9	7.4	2.388	7.570	+4.5	+5.182
Cubic BP	Z	21.7	> 36	21.908	unresolved	> 14.3	unresolved
Cubic BP	Dep.	4.3	4.2	4.223	4.117	-0.1	-0.106
Cubic Sweep	X	2.9	7.1	3.031	6.816	+4.2	+3.785
Cubic Sweep	Z	14.7	22.2	14.182	19.473	+7.5	+5.291
Cubic Sweep	Dep.	4.4	4.7	4.894	4.759	+0.3	-0.134
Rhombic BP	X	1.0	1.7	1.047	1.864	+0.7	+0.817
Rhombic BP	Z	28.4	6.5	27.407	6.490	-21.9	-20.916
Rhombic BP	Dep.	1.5	1.7	1.640	1.649	+0.2	+0.009

The source fits a quadratic finite-size form $y = A + Bx + Cx^2$, $x = (p - p_{\text{th}})L^{1/\nu}$, by unweighted least squares (source equations 3.1–3.2), following the scaling method discussed by Tuckett et al. (2020). We fit this form with bounded parameters and compare it with an exploratory normalized-x reparameterization that relaxes coefficient and exponent bounds. Fits drop individual probability endpoints and extreme sizes as diagnostics. The two parameterizations are reported separately because their bounds differ.

Each two-arm family has 400 conditional binomial bootstrap draws. Bounds, nonphysical predictions and grid extrapolation prevent robust inference in several families. The principal source-comparison family has 82 quantities: 24 BP families times three quantities, three Sweep families times three, and the undeformed cubic Z threshold. The pure-X threshold experiment has a separate three-comparison

family. We save unadjusted and Bonferroni-adjusted conditional bands; normal bands based on bootstrap standard errors are approximations, explicitly withheld as inference when diagnostics fail. Six cross-code contrasts form a separate family. These adjustments neither supply missing source uncertainty nor control finite-size/model bias.

Family	Arm	Ours − source (pp)	Conditional band (pp)	Rounding
cubic dep.	U	-0.077	[-0.66, +0.51]	close
cubic dep.	D	-0.083	[-0.57, +0.40]	close
rhombic dep.	U	+0.140	[-0.31, +0.59]	close
rhombic dep.	D	-0.051	[-0.38, +0.28]	close
rhombic X	U	+0.047	[-0.11, +0.20]	match
rhombic X	D	+0.164	[-0.77, +1.10]	mismatch
rhombic Z	U	-0.993	[-4.02, +2.03]	mismatch
rhombic Z	D	-0.010	[-0.92, +0.90]	match
sweep X	U	+0.131	[-0.75, +1.01]	close
sweep X	D	-0.284	[-1.61, +1.04]	mismatch
sweep Z	U	-0.518	[-1.63, +0.59]	mismatch
sweep Z	D	-2.727	[-4.75, -0.70]	mismatch
sweep dep.	U	+0.494	[-0.18, +1.17]	mismatch
sweep dep.	D	+0.059	[-0.51, +0.62]	close
cubic Z	U	+0.208	[-3.28, +3.70]	mismatch

The displayed bands concern ours-minus-source deviations, in percentage points. They exclude unknown source Monte Carlo error. Table rounding uses half-up 0.1 pp precision: a match has identical rounding, and close is one last-place unit. Rounding agreement is distinct from uncertainty overlap and limited reconstruction scope. Figure coordinates have a conservative 0.1 pp extraction allowance, not an asserted original precision or trial error. Difference contrasts accumulate both input allowances.

Undeformed cubic pure-Z has bootstrap percentile interval 20.884–24.996%, with four boundary fits among 400 draws; deleting an extreme size moves its estimate to 20.745–23.082%. The deformed curve saturates and its sector fits hit boundaries. A numerical polynomial output around 42% is therefore not reported as a resolved threshold. The source itself describes the 36–50% crossing as uncertain.

Rhombic pure-Z yields a signed deformation contrast −20.916 pp versus source −21.9 pp. Its bootstrap fits have no optimizer-bound hits in this grid, but unknown configuration and finite-size effects limit interpretation. The small rhombic depolarizing fitted advantage, +0.009 pp versus source +0.2 pp, does not establish a meaningful ranking. Sweep depolarizing changes from source +0.300 pp to ours −0.134 pp; seven bootstrap fits hit bounds, and modern matching further limits the comparison.

4 Finite bias and cross-code comparisons

Code/axis	η	Source (%)		Ours (%)		$D-U$: source/ours (pp)
		U	D	U	D	
Cubic X	1	3.52	4.12	3.70	4.15	+0.60/+0.45
Cubic X	3	3.33	4.70	3.11	4.73	+1.37/+1.61
Cubic X	10	3.04	5.68	2.96	5.61	+2.64/+2.65
Cubic X	30	2.86	6.75	2.97	6.27	+3.89/+3.30
Cubic X	100	2.89	6.81	2.93	6.94	+3.92/+4.02
Cubic Z	1	5.56	5.08	5.57	4.94	-0.48/-0.63
Cubic Z	3	12.12	7.25	11.66	8.50	-4.86/-3.16
Cubic Z	10	22.35	12.44	22.51	13.19	-9.92/-9.31
Cubic Z	30	21.74	16.00	21.53	19.71	-5.74/-1.82
Cubic Z	100	21.71	23.75	21.03	37.83	+2.04/+16.80
Rhombic X	1	1.27	1.44	1.45	1.64	+0.17/+0.20
Rhombic X	3	1.16	1.50	1.22	1.63	+0.34/+0.41
Rhombic X	10	1.09	1.47	1.21	1.70	+0.37/+0.49
Rhombic X	30	1.07	1.59	1.04	1.80	+0.52/+0.76
Rhombic X	100	1.00	1.63	1.67	1.87	+0.63/+0.20
Rhombic Z	1	2.09	1.80	2.15	1.84	-0.29/-0.31
Rhombic Z	3	4.00	2.47	4.59	2.59	-1.53/-1.99
Rhombic Z	10	11.14	3.90	11.95	3.86	-7.25/-8.08
Rhombic Z	30	28.94	5.11	28.52	5.53	-23.82/-23.00
Rhombic Z	100	28.00	6.00	29.93	5.67	-22.00/-24.26

Figure-derived source values above retain two decimal places only for readability; their extraction allowance is 0.1 pp. These are all plotted finite-bias conditions, with both arms and all four lattice sizes. The source methods list $\eta = 5$ but vector plot coordinates put the corresponding point at 3. We run both; only 3 is compared with that plotted coordinate. The eta-5 results evaluate the stated method; no corresponding plotted source value is available.

An exploratory design broadened the three grids whose fits were unstable: cubic Z eta-5 undeformed gives 18.905% but nonphysical/nonmonotone polynomial predictions; cubic Z eta-100 deformed reaches the 49.99% optimization bound outside the sampled grid; rhombic X eta-100 undeformed gives 0.725% with some negative fitted probabilities. None resolves the affected original estimate simply by adding shots over a broader range.

The source ranks undeformed rhombic highest at Z bias 30–100. Five of six reconstructed pairwise point contrasts retain that direction. At eta-100, undeformed rhombic minus deformed cubic is -7.898 pp, reversing the source point ordering. The cubic fit is not identified reliably, including in the wider-grid check; this is evidence that the ranking is not reproduced robustly, not evidence for a new asymptotic winner. No source ranking is statistically established without its uncertainty.

5 Finite-size scaling diagnostics

Neither the logistic nor the quadratic scaling model adequately describes the tested local grids. To assess scaling separately from the broad-grid estimates, we selected grids using the broad-grid crossing brackets, without source thresholds as inputs, and froze them before generating 112,000 trials at all four sizes. The two local families use 1,000 trials per generator and condition; the broad sector study uses 500. The 144 generator/seed pairs are disjoint from the broad-grid samples. Largest-size pilots used 60 trials each and cost 3.170, 0.364 and 2.460 seconds; two-worker production took 722.614 seconds, with peak resident memory below 70 MiB per process. Every BP correction was syndrome-valid. These diagnostic samples are analyzed separately from the threshold-estimation samples.

Family	Local quadratic (%)	Logistic (%)	Deviance/16 df
Cubic Z, $\eta = 5$, U	20.5414	20.5725	137.219
Rhombic X, $\eta = 100$, U	1.05866	1.08223	56.699

These are diagnostic fitted outputs, not accepted thresholds. The quadratic curves are physical and monotone on these frozen grids, unlike the broad-grid fits, but that alone does not establish scaling. The independently fitted bounded model is $\mathrm{logit}(P) = a + b[(p - t)/\,\mathrm{median}(p)](L/6)^{1/\nu}$. Its approximate goodness-of-fit probabilities are 2.52×10^{-21} and 1.86×10^{-6}: both reject this model. Its profile intervals therefore are not reliable threshold intervals. A parametric-bootstrap adequacy test evaluates the quadratic estimator on the same fixed grids. Counts are drawn from each fitted quadratic curve, and the identical five-start unweighted estimator is refitted. Observed binomial deviances are 138.190 and 41.372; none of 2,000 refits per family exceeds them. The add-one tail estimate is 0.000500, with Monte Carlo 95% upper bound 0.001843 for each. All 4,000 refits pass the declared validity checks; failed or nonphysical fits would have bounded, not disappeared from, the tail calculation. This rejects both quadratic models at the two-model Bonferroni level and the stated 0.005 sensitivity. It is an exploratory plug-in diagnostic, not an exact finite-sample test or threshold interval. Production took 53.6 seconds after excluded pilots; no physical-error trials or windows changed. These grids do not support an established threshold under either tested model; inference requires a justified alternative model and a prospective size/grid design. Fits omitting the smallest or largest size are reported as sensitivity analyses, not as preferred estimates. There is no unique plotted eta-5 source value because of the methods/figure discrepancy.

For cubic eta-5, the $L = 4/6$, $4/8$ and $4/10$ interpolated crossings are 23.0704%, 21.7571% and 20.3394%; the remaining pairs fail the predeclared uniqueness/stability guard. Corresponding rhombic crossings are 1.16486%, 1.12861% and 1.07830%, with the other pairs likewise failing. A 12-member Bonferroni bootstrap family covers finite-grid interpolation conditionally on a unique root; it is not asymptotic threshold coverage. Missing, multiple or oppositely oriented crossings are retained, not fitted away.

Deformed cubic Z at eta-100 illustrates separate sector changes. At $L = 10$, total/point/loop failure is 0.494/0.494/0 at $p = .20$, 0.521/0.513/0.018 at .28, and 0.887/0.513/0.764 at .40. Eleven of 72 adjacent-size/sector signed differences exclude zero under conservative simultaneous exact-binomial intervals. All 72 comparisons are reported. No single global threshold is forced onto these curves, and the descriptive separation is not a causal or asymptotic proof. Point and loop failures share trials. Separate scalar likelihood and raw-count aggregation checks reproduce the analysis; they share the observations and model definitions, not an independent physical simulation.

6 Full curve and analytical coverage

The four pure-channel/code combinations cover total, point and loop failure for both arms and every size. The source supplies 6,804 factual vector coordinates across 96 curves. The reconstruction samples their horizontally displayed probability vertices, snapping calibrated coordinates to the source's 0.005 grid within extraction tolerance. Below-axis source values are treated as bounds rather than exact zero observations. The comparison uses a declared 6,804-member Bonferroni Clopper–Pearson family for our binomial counts plus a 0.005 vertical extraction allowance; source trial counts are unknown. All 6,804 source-coordinate intervals overlap our simultaneous bands, with zero invalid BP corrections in these 226,800 trials. This is a low-resolution compatibility check: only 100 fresh trials per condition and a large multiplicity family produce wide bands. It is not numerical equivalence, a recovered source trial archive, or evidence that every plotted value was exactly reproduced. The source has 4,322 visible vertices; the remaining below-axis coordinates are treated conservatively as probability intervals [0,0.015].

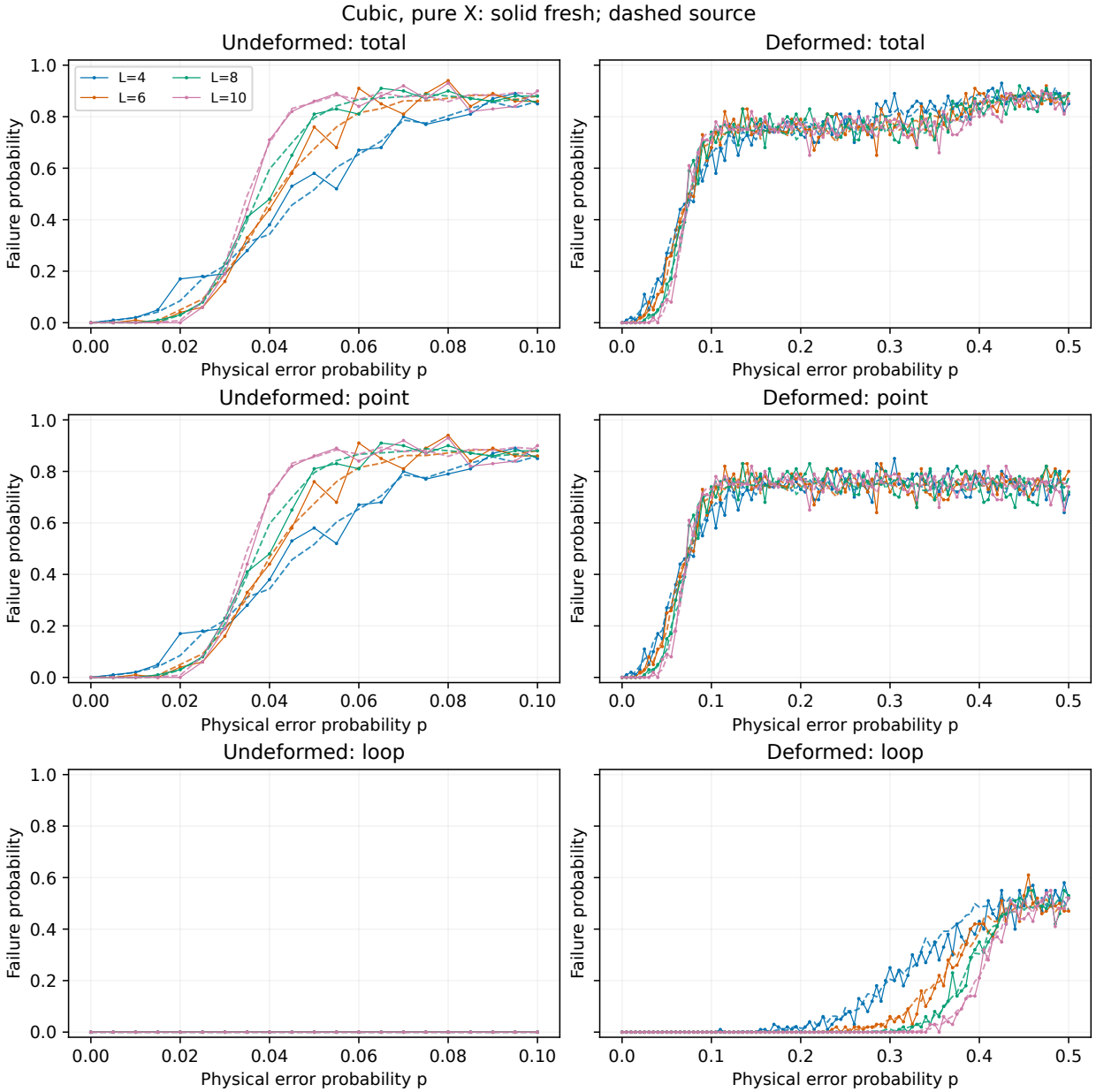


Figure 1: Cubic pure X: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

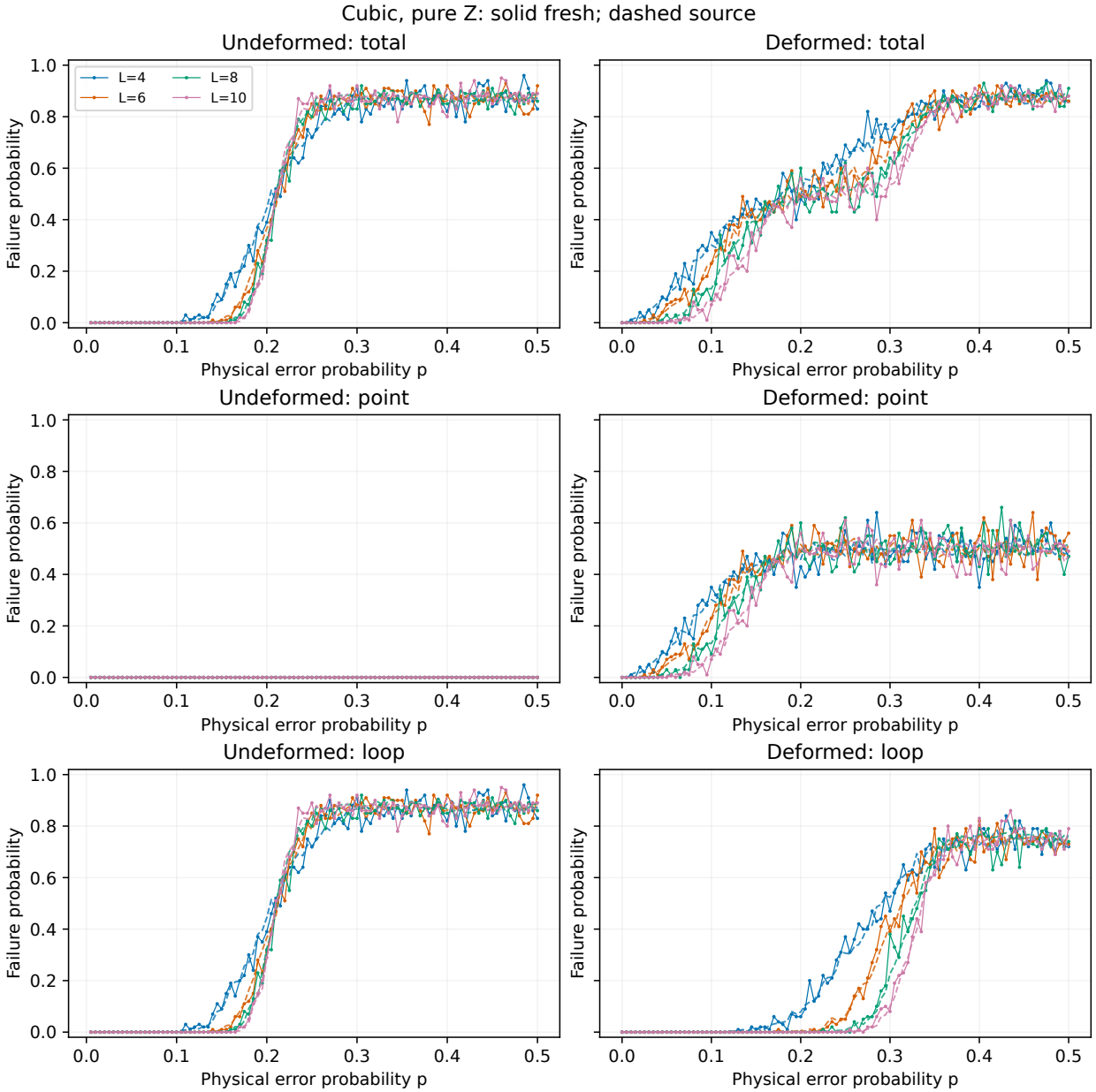


Figure 2: Cubic pure Z: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

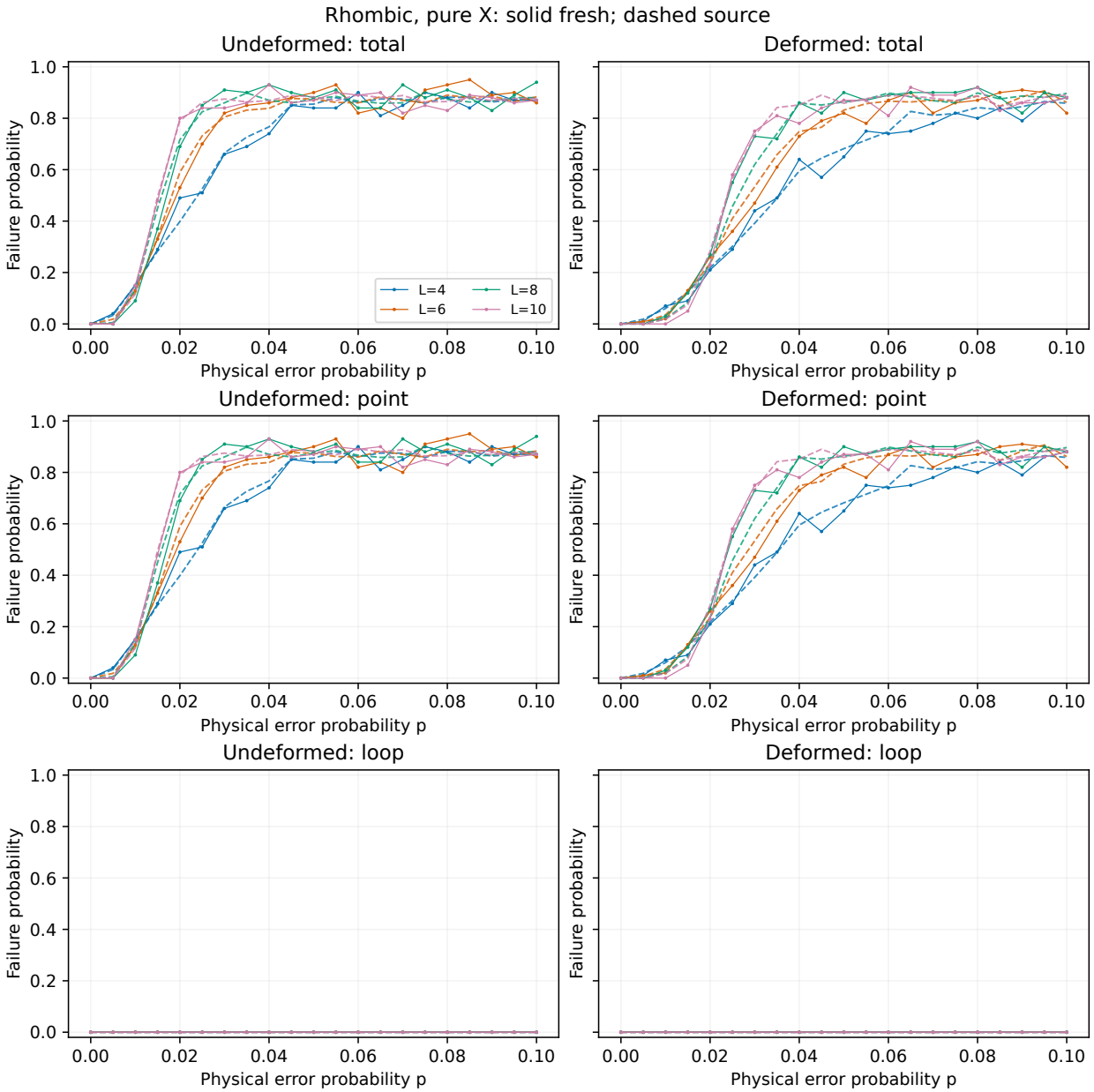


Figure 3: Rhombic pure X: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

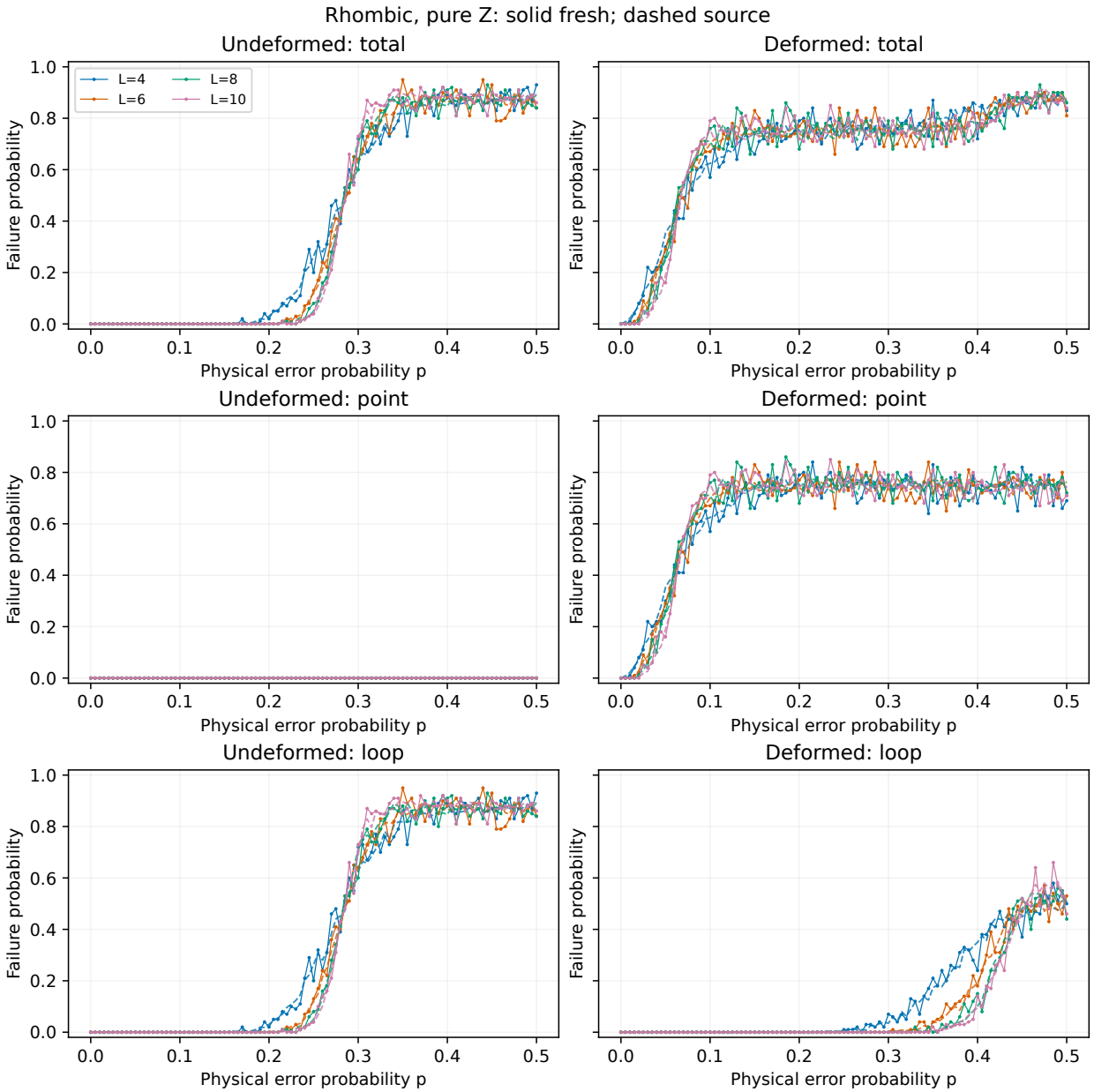


Figure 4: Rhombic pure Z: all sizes, arms and sectors. Solid lines are fresh trial frequencies; dashed lines are source vector coordinates. Connected lines guide the eye; no pointwise error bars imply simultaneous precision. Below-axis source coordinates are not exact observed zeros.

Finite geometry checks establish commutation, three logical qubits, matched logical operators and restricted-axis nullities at $L = 2, 4, 6, 8, 10$. Computational distance certificates verify matching lower and upper bounds at these sizes: cubic $d_X = L$, $d_Z = L^2$, and rhombic $d_X = L$, $d_Z = 2L^2$ under our convention. The lower-bound check uses disjoint homologous representatives and exact row-space membership. These are finite computational certificates, not Lean proofs or an arbitrary-L theorem.

For the ideal repetition-chain and plane reductions, a separate analytical audit evaluates finite binomial majority probabilities and parity composition, with explicit independent-component and random-tie assumptions. The 50% ideal-sector limit is not the threshold of the actual BP–OSD decoder, nor does it establish the total-code threshold. The rhombic pure-X loop claim is checked through finite restricted-axis nullities. Lean 4.22.0 verifies general binary Hadamard preservation, restricted difference kernels and the cubic cohomology and cocycle-distance results described below. Appendix B states the exact correspondence between formal definitions and source objects. Section 7 supplies the written quantum, rhombic and probability arguments, distinguished from Lean coverage.

6.1 General cubic cohomology and distance

The source states $k = 3$, $d_X = L$ and $d_Z = L^2$ for the periodic cubic construction (Section 2.4.1, pp.28–29). Our general Lean result classifies closed periodic binary cochains by exactly three Boolean cycle labels, modulo vertex gradients, and proves minimum nontrivial cocycle weight L^2. A finite-coordinate bridge connects zero holonomy to the actual vertex-gradient formula, including wraparound. The general theorem establishes the coordinate-model Z-distance; the interpretation as a quantum code also uses the written argument in Section 7.

Source claim	Reconstructed consequence	Evidence class
Three logical pairs	Three independent cycle labels	General coordinate Lean classification
Z distance L^2	Minimum nonzero cocycle weight L^2	General Lean lower bound and witness
X distance L	Dual cycle/slab argument	Written duality proof
$k = 3$	Vertex/face ranks $L^3 - 1$, $2L^3 - 2$	Written rank consequence

A separate consumer checks all eight labels at odd and even sizes $L = 2, 3, 4, 5, 6$, using two input potentials: 80 fields and 21,120 gradient entries agree with the Python matrices. It checks nonzero period jumps rather than incorrectly assuming every closed field has a periodic potential. These constructed examples share inputs and geometric conventions; they validate implementation connections, not the general theorem by sampling. Appendix C gives the constructive proof and its scope.

7 Code parameters and decoder guarantees

The code parameters, ideal-majority endpoint and conditional OSD guarantee are analytically verified; the actual BP–OSD asymptotic endpoint does not follow from these results. Table 3 separates these outcomes. The complete arguments and assumptions are in `analytical-completion.md`; these are written proofs, distinct from the kernel-checked statements above. The written arguments use standard torus homology, finite binary rank-nullity and Pauli trace identities.

Source statement	Verified result	Evidence and boundary
Rhombic parameters, pp30–32	$n = 3L^3$, $k = 3$, $d_X = L$, $d_Z = 2L^2$ for even $L \geq 2$	Written FCC cellulation and disjoint-cycle proof; independent matrices at five sizes. Prose Pauli convention.
Cubic reductions, p44	2D point layers; ideal chain/plane majority endpoint 50%	Restricted-kernel proofs plus written probability argument; 48 exact-rational comparisons. Not the actual BP threshold.
Rhombic loop, pp55–56	Unique recovery on first-axis support	Restricted formula proof. Four outside-grid stress failures qualify unrestricted implementation wording.
OSD, pp34–40	Correct syndrome when $s \in \operatorname{im} H$	Written elimination invariant; 20,748 exhaustive cases across three solvers. No optimality or BP convergence claim.
Hadamard, pp25,44	n, k, d preserved	Binary Lean transport plus written unitary/projector bridge. Mixed-code sector distances not asserted invariant.

Table 3: Analytical statements from Pesah (2021), with our verification and its limits. Written arguments and finite checks are not relabeled as Lean proofs.

7.1 Rhombic homology and distance

The odd-parity cubes define vertices of a face-centred cubic complex. Each physical edge touches exactly two such cubes and becomes their connecting edge. The four odd cubes around an original vertex form a tetrahedron: its six edges are the six incident physical edges, and its four faces are the source triangle checks. Cube and triangle incidence therefore commute. Subdivide each unit cube into its odd-corner tetrahedron and four even-corner pieces. The latter join into octahedra. This fills space; even periodic identification gives a three-torus, including small quotients with identified edge endpoints.

There are $L^3/2$ vertices and $3L^3$ edges. Connectivity gives cube rank $L^3/2 - 1$; three-dimensional first homology gives triangle rank $5L^3/2 - 2$, hence $k = 3$. A nontrivial cycle has a trail with displacement at least L in one coordinate, while each edge changes that coordinate by at most one. A length-L zigzag attains the resulting X-distance. For each winding direction there are $2L^2$ edge-disjoint homologous zigzags, one for each transverse position and each of two physical edge orientations. A nontrivial cocycle intersects all of them oddly, so its weight is at least $2L^2$. A transverse cut with both orientations attains this Z-distance. The source's prose and figure/historical Pauli labels conflict; these formulas follow the prose convention.

`check_rhombic_cellulation.py` reconstructs endpoints and tetrahedron faces independently of the source corner-triple implementation. At $L = 2, 4, 6, 8, 10$, every cube matrix and triangle multiset matches, and all counted zigzags have connected endpoints, correct winding and disjoint supports. The check takes 0.289 seconds and under 84 MiB RSS. Its shared periodic-coordinate convention limits independence; finite cases support the mapping, while the general result uses the written cellulation proof.

7.2 Ideal majority and actual decoding

For a component of m independent Bernoulli(p) errors, let $q_m = \Pr[B > m/2] + \frac{1}{2}\Pr[B = m/2]$, where $B \sim \mathrm{Bin}(m, p)$ and ties use independent fair coins. Parallel components represent the same logical class,

so the failure probability for N components is

$$P_{\text{fail}} = \frac{1-(1-2q_m)^N}{2}.$$

The restricted cubic sectors have $(m, N) = (L, L^2)$ for chains and (L^2, L) for planes. For fixed $p < 1/2$, $q_m \le \exp[-2m(1/2-p)^2]$ and $P_{\text{fail}} \le Nq_m \to 0$. One direct derivation uses exponential Markov inequality: for $0 < p < 1/2$, choose $t = \log[(1-p)/p]$ to bound $\Pr[B \ge m/2]$ by $[2\sqrt{p(1-p)}]^m \le \exp[-2m(1/2-p)^2]$; $p = 0$ is immediate. At $p = 1/2$ the logical parity is uniform. This proves the ideal-majority endpoint on $0 \le p \le 1/2$, not that ten-iteration BP–OSD realizes it. The remaining point equations at fixed first coordinate are exactly square-lattice vertex incidence; `check_cubic_layers.py` independently verifies the finite mapping.

`check_majority_exact.py` independently evaluates Bernoulli polynomial convolution and parity recursion using rational arithmetic. All 48 saved predictions at four sizes and six probabilities agree within 1.43×10^{-15}; the 2×10^{-14} check tolerance concerns floating-point evaluation, not sampling uncertainty. The argument requires independent errors and independent tie coins. Further finite-size fits cannot turn this ideal-decoder proof into a BP threshold theorem.

In the deformed rhombic loop sector, each first-axis error is exposed by a triangle, proving unique recovery restricted to that support. An unrestricted regularized decoder can nevertheless leave it. At $L = 2$, exhaustive tests of 256 patterns at prior 10^{-12} find four valid-syndrome logical failures; the external historical dense implementation reproduces all four sparse-decoder corrections. At priors $.001, .1, .49, .5$ every pattern succeeds. `check_loop_reference.py` and `rhombic-loop-exhaustive.json` retain the witnesses. The source wording is preserved in `rhombic_loop_argument.png` and `rhombic_loop_continuation.png`. These stress cases lie outside the dissertation's sizes and probability grids: they qualify a universal implementation guarantee, not its reported zero-loop-error observations or the restricted uniqueness theorem.

7.3 OSD and the quantum bridge

Column permutation and invertible row operations preserve the system $Hc = s$. If $s \in \operatorname{im} H$, the zero rows after elimination have zero right-hand side. Setting free variables to zero and pivot variables to their reduced right-hand side solves every equation; undoing the permutation yields a valid correction. The native solver's skipped prefix words are zero by its active-row invariant. The source/Python solvers omit an inconsistency guard, so the consistent-syndrome premise is essential; all production syndromes here satisfy it.

An independent column-span enumeration in `check_osd_exhaustive.py` checks all 682 binary matrices of shapes 1–3 by 1–3, every consistent syndrome, all strict reliability orders and a tied order. All 20,748 cases match Python, native and historical solvers: 62,244 correction comparisons. In 455 inconsistent examples native rejects while source/Python return invalid corrections. The complete check takes 3.34 seconds and under 48 MiB RSS. Shared matrices and tie ordering are disclosed. This verifies consistency, not optimality or convergence of approximate BP.

For r independent commuting Hermitian Pauli checks generating no $-I$, the common-eigenspace projector is $2^{-r}\prod_i(I+g_i)$. Only its identity term has nonzero trace, giving dimension 2^{n-r}. Independent CSS bases satisfy the no-$-I$ premise because a scalar product requires both binary supports to vanish. Tensor-product Hadamard conjugation is unitary and preserves Pauli support, including the irrelevant sign on Y. This supplies the Hilbert-space interpretation of the binary transport theorem; it does not assert unchanged CSS sector distances after mixing.

7.4 Limits of the analytical guarantees

The ideal-majority 50% endpoint does not prove an asymptotic threshold for BP–OSD. General cubic rank/X-distance, rhombic homology/distances and quantum interpretation are established by written arguments; they are not presented as Lean theorems. The frozen numerical diagnostics reject the tested local scaling models; changing windows to obtain a preferred threshold is not justified. The unavailable inputs are original production counts/configuration, benchmark inputs and the frontend dependency. Circuit-noise and other Chapter 4 proposals are future work rather than completed dissertation experiments.

8 Software performance and interface reproducibility

The source reports a tenfold OSD speed improvement from reusing elimination rather than recomputing ranks. Our reconstructed repeated-rank baseline and reused-elimination implementation agree on 15 small cases and show speedups roughly 12–145 times. The original benchmark inputs, baseline implementation and hardware are absent; concurrent workload also affects our timings. This supports the mechanism, not an exact tenfold benchmark reproduction. The native optimization's separate speedup is not credited as a reproduction of that source claim.

On shared diagnostic errors, total failures are 1,375 at ten iterations, 1,476 at 100, and 1,353 with stable sorting, out of 5,600 each. These heterogeneous exploratory counts demonstrate setting sensitivity; they are not pooled threshold estimates. No invalid correction occurred in these checks.

The historical Flask backend passes 28 response checks: four code/size geometry checks and 24 zero-syndrome BP decoding requests. Browser loading of the original frontend encounters a denied external Three.js module host and produces no canvas. Frontend interaction and rendering could not be tested because the required module was inaccessible. Backend success is not full GUI reproduction. Claimed usefulness for visualization is qualitative and lacks a source evaluation protocol.

Chapter 4 proposes measurement/circuit noise, coprime dimensions, boundaries, higher-order OSD, a statistical-physics model and future resource comparisons. They are future work, not dissertation experiments that we failed to run. The specification of general Pauli channels does not supply experimental results for separate Y-bias or circuit-noise studies.

9 Evidence availability and methodological checks

The full dissertation and historical third-party code are obtained externally. Limited attributed evidence crops, vector factual coordinates and original reconstruction artifacts are governed by the repository's licensing notices. Source evidence: `table3_1.png`, `scaling_equations.png`, `table3_2.png`, `cubic_bias_figure.png`, `rhombic_bias_figure.png`, `methods_grid.png`, `manifest.json`, `expansion-manifest.json`. The tables summarize artifact availability and methodological checks; detailed records are `expanded-availability.md` and `expanded-compliance.md`.

Item	Result and limit
Thesis and production archive	Complete 66-page author-hosted MRes source obtained at b4f401e0c12b44be4c69318e0015f70d8a3d3f98. All 47 returned author repositories and historical collaboration code searched; no pinned production trials/settings archive identified. The search scope does not establish exhaustive absence.
Historical decoder source	PanQEC revision 686560e5d03c2e5d31f6c6260b03756af03a7e93 obtained externally. Dense BP and Sweep routines executed without vendoring. Exact dissertation production identity unknown.
Historical matching dependency	PyMatching0.2.4 could not be built: original setup fetched a LEMON archive from lemon.cs.elte.hu and received HTTP403. PyMatching2.3.1 is a labeled reconstruction, not historical equivalence.
Graphical interface	Historical Flask backend tested. Original frontend requires cdn.skypack.dev/three@v0.130.1; CONNECT403 prevented module access and canvas rendering.
Unspecified research inputs	Original trial counts, seeds, BP settings, exact package pins, fitting windows and OSD benchmark workload/hardware were not found. Source uncertainty is unknown.
Permitted evidence	Eight attributed 600-dpi crops and factual vector coordinates retained with provenance; complete thesis and third-party source remain external. THIRD_PARTY.md and LICENSING.md apply.

Item	Result and limit
Study scope	87-item inventory of numerical, analytical, software and qualitative results, with background and future work distinguished. The source is a UCL Master of Research dissertation.
Decoder implementations	Binary BP/OSD and Sweep–Match implement the source coding problems. Python, native and historical-code comparisons test correction consistency; PyMatching 2.3.1 is a disclosed substitute for the unavailable historical matching build.
Numerical coverage	All endpoint cells, plotted finite biases, the eta=5 method alternative and pure-channel sector/arm/size curves are assessed. The threshold and curve experiments contain 472,400 trials; cubic pure-X uses a separate 16,000-trial sample. Unstable fits are not accepted thresholds.
Independent implementations	480 cubic and 320 rhombic historical dense correction matches; 196 native/Python OSD cases through L=10; 80 matching/exhaustive-pairing cases. Shared inputs, graphs, priors and software limit independence.
Resources and defaults	Largest-size pure-Z pilots cost approximately 8.43/9.10 seconds per ten trials; the 8,800-trial run took 2,478 seconds. Processes are bounded by 2,400 CPU seconds and 2 GiB address space. Ten BP iterations, order-zero OSD and tie handling are supplied settings.

Item	Result and limit
Contrasts and precision	D minus U in percentage points favors deformation when positive. Ours-minus-source deviations are separate. Table rounding, figure extraction, source uncertainty and reconstruction scope are distinct; point reversals have model/engine qualifications.
Uncertainty families	82 threshold/source quantities, six cross-code contrasts and 6,804 curve-coordinate checks use separate alpha=0.05 Bonferroni families; cubic pure-X has a separate three-comparison family. Conditional bootstrap-normal bands are not reliable when fit diagnostics fail. Source Monte Carlo error is unknown.
Sampling and failures	Distinct seeds within each run; 406 generator/seed keys overlap separately reported families, which are not pooled independently. Cross-code seeds are disjoint. All 2,696 residual-syndrome Sweep trials count as failures; BP production corrections are syndrome-valid.
Scaling adequacy	112,000 diagnostic trials on frozen grids, zero invalid corrections, 722.614 seconds with two workers. Twelve crossing and 72 sector-contrast families. Logistic models and both source-family quadratic models fail adequacy checks; 4,000 quadratic refits used fixed physical samples and windows.
Formal verification	Lean 4.22 verifies Hadamard transport, cubic incidence and restricted kernels, three-label cohomology and cocycle distance. 609,792 cross-language matrix entries match; 80 consumer fields and21,120 gradient entries are checked. Standard axioms include Classical.choice; no added axioms or admitted proofs.
Written verification	General cubic rank/X-distance, rhombic rank/distances, quantum interpretation, ideal-majority limits and conditional OSD consistency are written arguments. 48 rational probability comparisons and 20,748 exhaustive OSD cases support implementation checks; these are not additional Lean theorems.
Limits of interpretation	Ideal-majority results do not prove an actual BP-OSD asymptotic endpoint. Four outside-grid loop witnesses qualify unrestricted implementation wording. Missing production/benchmark inputs and inaccessible matching/frontend dependencies prevent exact historical equivalence.
Reporting and reproducibility	Headline numbers and limits, full-condition tables, attributed source evidence and availability rows are included. Immutable reproduction instructions distinguish saved-output analysis, fresh simulation and evidence extraction.

10 Conclusion

We did not establish the reported cubic pure-Z improvement from 21.7% to greater than 36%. Our undeformed estimate is 21.908%, but the deformed threshold is not identifiable reliably from these finite-size curves. In contrast, the reconstructed cubic pure-X and rhombic pure-Z comparisons recover the published directions: deformation improves the former and worsens the latter. The Sweep–Match depolarizing contrast reverses sign, and the cross-code ranking at Z bias 100 is sensitive to an unstable cubic

fit. Unknown source uncertainty and production settings prevent these discrepancies from establishing a statistically supported opposite conclusion.

Analytical verification explains several structural properties without resolving the missing performance result. The cubic and rhombic parameters, ideal-majority limits and conditional OSD guarantee follow from the stated proofs and assumptions. Actual BP–OSD asymptotic thresholds require additional justification. Reliable numerical threshold estimates require an adequate scaling model and a prospective size/grid design; exact production and benchmark reproduction require the original inputs and configuration. The unavailable matching dependency and frontend module also prevent historical-engine and full-interface equivalence. These limits constrain the replication verdict rather than weakening the distinction between a reported value, a reconstructed estimate and a verified theorem.

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A Reproduction artifacts

The complete numerical and analytical reproduction code, saved outputs and prerequisites are available at immutable revision `ed52777cee600ce33e24d45a43f8938d8cc5d15d`: `expansion/README.md` describes the experiments and `followup/README.md` describes the proof and diagnostic checks. The historical dissertation and reference code are external prerequisites with pinned revisions and hashes. Regenerating tables from saved outputs, rerunning simulations and extracting source evidence are separate commands in those instructions.

The computations use Python 3.12.14 for the runtime and integer binary elimination, NumPy 2.3.5 for arrays and sampling, and SciPy 1.17.0 for sparse arithmetic and fitting (Python Software Foundation, 2026; NumPy developers, 2025; SciPy developers, 2026). Historical reference validation uses qecsim 1.0b9 with its NetworkX 2.6.2 dependency (Tuckett, 2021; NetworkX developers, 2021). Matplotlib 3.10.8 produces the scientific figures; Pillow 12.3.0 processes raster evidence (Matplotlib development team, 2025; Pillow contributors, 2026). These are the reconstruction environment versions, not an assertion of the source's production environment.

The accompanying editable manuscript and bibliography are `expanded-manuscript.tex` and `expanded-references.bib`; `expanded-build.sh` builds the main PDF. Evidence and comparison data are linked in the relevant sections. The scientific code and evidence links identify immutable repository snapshots.

B General algebra: proof-to-source correspondence

Lean 4.22.0 with Std verifies the statements below without admitted proofs or added axioms. Standard dependencies are `propext`, `Quot.sound` and, for existence and classification, `Classical.choice`. The executable build is `build.sh`; the exact source-to-formal correspondence is documented in `cubic-proof-bridge.md`. These are formal reconstructions of standard code properties, not mathematical novelty claims.

Source obligation	Exact checked statement	Interpretation and scope
Hadamard, 2.3.3/3.2	`binary_parameters_preserved`: arbitrary finite qubits and swap mask preserve isotropic subgroup, basis-witness rank and minimum logical weight.	Binary Pauli semantics modulo phase; Section 7 supplies the written Hilbert-space interpretation.
Cubic chains, 3.2.1	`chain_kernel`: periodic first-axis differences vanish iff constant along each chain.	Connected to the six-edge vertex formula in Lean and checked against finite matrices. Written counting and ideal-majority limit are in Section 7.
Cubic planes, 3.4	`plane_kernel`: periodic y/z differences vanish iff constant on each yz-plane.	Connected to four-edge face formulas in Lean. Written counting/probability in Section 7; no BP threshold implication.

Source obligation	Exact checked statement	Interpretation and scope
Rhombic triangles, 3.5.2	`actual_two_corner_family_injective`: two restricted corner orientations expose every first-axis coordinate.	Formula-to-code correspondence is manually checked; Section 7 verifies the full even-periodic geometry and specifies the Pauli convention.

A binary Pauli is a function $\{0, \ldots, n-1\} \to \mathrm{F}_2^2$. The deformation swaps its components on a chosen mask. The proof verifies involution, XOR linearity, support and full-qubit symplectic-product preservation. It then transports subgroup membership, normalizers, logical cosets, a generating independent list and exact minimum-weight witnesses. The definition of encoded qubits is the explicit basis-rank relation $n = r + k$, not an independently proved Hilbert-space dimension theorem. `first_axis_parameters` instantiates $n = 3L^3$ and the mask $i < L^3$, matching the Python first-third swap. No CSS sector-distance invariance is asserted after mixed deformation.

The periodic kernel proof uses induction around one coordinate: equality of adjacent bits implies constancy; applying it twice gives constant planes. In the cubic implementation, vertex restriction supplies neighboring first-axis edges (with a cyclic index shift), while the two face orientations supply y/z differences. The formal difference model allows positive lengths; the actual source lattice is used only at its stated admissible sizes, not a degenerate length-one implementation.

For rhombic triangles, the source reconstruction has one edge per axis and first-axis inward bit equal to base-X XOR vertex parity. Under first-axis-only error the other terms vanish. Orientations 0 and 2 supply base-X bits 0 and 1. Choosing base-X equal to vertex parity exposes the unshifted edge at every vertex, proving restricted-formula injectivity. The theorem concerns the explicit corner formula; correspondence with the Python incidence matrix is checked separately. The full code requires even periodic sizes; no odd-size physical result is claimed.

The statements in `CubicIncidence.lean` define all six vertex terms and all three four-edge face orientations on periodic coordinate triples. `cubic_incidence_commutes` proves that faces annihilate arbitrary vertex gradients: each corner bit occurs twice. `vertex_gradient_kernel` proves that zero gradient is equivalent to global constancy. The restricted chain and plane kernels are connected to these formulas, including the periodic backward/forward inverse. `chain_syndrome_classification` characterizes equal restricted vertex syndromes by chain-constant XOR differences; component parameters are unique. These are general formula proofs, not numerical extrapolations.

A separate Lean consumer exports incidence entries, which are compared with the Python construction: six matrices at $L = 2, 4, 6$, 609,792 entries and zero mismatches. Implementations share geometric conventions but not the matrix generator. This is an exact finite cross-language bridge; arbitrary-size array flattening is checked manually rather than proved in Lean. The abstract XOR formulas allow positive lengths, whereas concrete matrix claims exclude degenerate length one. No general rank-number, logical-dimension or distance claim follows without its additional counting/homology proof.

C Constructive periodic classification and distance

The constructive proof classifies closed periodic cochains by their three cycle labels and establishes their minimum nontrivial weight. Its statements are in `CubicPotential.lean`, `CubicDistance.lean` and `CubicLift.lean`; `cubic-homology.md` gives the written argument and implementation correspondence. The Lean build and standard axiom dependencies are specified in Appendix B.

For a closed binary edge field on the unwrapped lattice, integrate along x at $y = z = 0$, then y at $z = 0$,

then z. A proved XOR telescoping identity and the three plaquette equations show that the resulting potential has the original field as its gradient. This potential is not assumed periodic. Period shifts change it by position-independent bits, precisely the three cycle sums. A periodic potential exists if and only if those bits vanish. Two closed fields therefore differ by a periodic gradient if and only if their labels agree. Seam representatives realize every label; constant fields would fail to realize nonzero labels at even L. The finite adapter proves the analogous zero-holonomy criterion for the actual finite-coordinate face and vertex formulas, including the periodic boundary.

A nonzero axis cycle sum is the same on all L^2 parallel lines. Each must contain an occupied edge, giving a kernel-checked lower bound L^2. A seam representative attains it. This is the exact minimum nontrivial cocycle weight in the coordinate model. The independently executable consumer uses actual Python vertex/face matrices and logical strings to check Lean-computed potentials, labels, weights and period jumps; all 80 constructed fields pass. General claims rely on proofs, not the number of examples.

For the written rank consequence, the constant vertex-gradient kernel gives rank L^3-1. The linear cycle-label map is onto three bits and its kernel consists exactly of gradients, so the closed-field dimension is L^3+2. Rank-nullity gives face rank $2L^3-2$, hence the CSS count $3L^3-(L^3-1)-(2L^3-2)=3$. Finite-dimensional rank-nullity/cardinality conversion, array-index rank correspondence and Hilbert-space interpretation are not all formalized; this is a written consequence of the checked structural results, not a Lean matrix-rank theorem.

For the written X-distance argument, finite binary annihilator duality pairs a non-boundary cycle with a nonzero seam class. Translated parallel seams differ by a vertex gradient, so a zero-syndrome cycle pairs identically with all L disjoint seams. Its weight is at least L; a straight periodic string attains L. This is a written analytical check, not a Lean duality theorem. The main analytical section supplies separate rhombic, probability, algorithmic and quantum arguments; no cubic result is transferred to the rhombic code without its own geometry.